Recovering Latent Distance-Deterrence Parameters from Aggregate Travel-Distance Distributions for Survey-Free Spatial Interaction Modeling

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Abstract

Accurate origin–destination (OD) flow estimation is fundamental to transportation planning, accessibility analysis, infrastructure assessment, and urban decision-making. Gravity-model calibration requires identifying city-specific distance-deterrence parameters from observed OD interactions. Although privacy-preserving aggregate mobility data are increasingly available, it remains unknown whether aggregate travel-distance distributions preserve sufficient statistical information for parameter identification without observing OD interactions. This study investigates whether latent distance-deterrence parameters can be identified directly from aggregate travel-distance distributions without observed OD interactions. Aggregate travel-distance distributions are extracted from Meta Movement Distribution Maps and formulated within a multinomial likelihood framework to infer latent distance-deterrence parameters. The recovered parameters are subsequently used to calibrate a gravity model for OD estimation. The inferred parameters closely match locally calibrated references and produce statistically comparable OD prediction performance across multiple metropolitan areas without requiring target-city OD observations. The findings demonstrate that aggregate travel-distance distributions preserve sufficient statistical information to identify latent city-specific distance-deterrence parameters. Building on this finding, the proposed probabilistic calibration framework enables transferable and privacy-preserving gravity-model calibration for data-scarce cities using only aggregate mobility observations.

Keywords: Spatial interaction, distance deterrence, gravity models, aggregate mobility data, parameter identification, zero-shot calibration, privacy-preserving mobility

1 Introduction

1.1 Background

Accurate origin–destination (OD) flow estimation is fundamental to transportation planning, accessibility analysis, infrastructure assessment, and urban decision-making (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Spatial interaction modeling represents the core analytical engine for estimating population mobility patterns across urban spaces. Gravity models (Wilson 1971; Tanner 1961) provide one of the most widely used probabilistic formulations of spatial interaction because they explicitly characterize how travel demand emerges from the interplay between origin production, destination attractiveness, and distance deterrence. Their predictive capability, however, depends critically on identifying appropriate city-specific distance-deterrence parameters.

1.2 Observation Challenge

The fundamental bottleneck in spatial interaction modeling lies in observation rather than model formulation. Identifying distance-deterrence parameters traditionally requires observing complete origin–destination flow matrices or individual mobility trajectories. However, comprehensive travel surveys are expensive and conducted infrequently (Egu and Bonnel 2020; Hussain et al. 2021), while trajectory records are tightly restricted by data privacy regulations and commercial licensing (de Montjoye et al. 2013). Consequently, conventional calibration methods fail in data-scarce urban regions worldwide because the necessary origin–destination observations are unavailable.

1.3 Opportunity from Aggregate Mobility

Aggregate mobility products—including Meta's Movement Distribution Maps (MDM) (Meta 2026)—are becoming increasingly available across global platforms and regions, providing privacy-preserving summaries of population travel patterns without revealing origin–destination pairs. Aggregate travel-distance distributions remove origin–destination identities while preserving the statistical distribution of travel distances across an urban area, suggesting a fundamentally different observation layer for spatial interaction systems.

1.4 Knowledge Gap

Existing studies have demonstrated that aggregate mobility products are valuable for descriptive travel analysis, accessibility assessment, and epidemiological monitoring (Oliver et al. 2020; Buckee et al. 2020; Pappalardo et al. 2023). Whether aggregate travel-distance distributions preserve sufficient statistical information for identifying latent spatial interaction parameters without observing origin–destination interactions, however, has not been established. Resolving this fundamental observation gap forms the central focus of this study.

1.5 Central Hypothesis

This study is founded on the following central scientific hypothesis:

> *Aggregate travel-distance distributions preserve sufficient statistical information to identify latent city-specific distance-deterrence parameters.*

1.6 Three Predictions

To systematically evaluate this hypothesis, we construct a sequential chain of three testable scientific predictions:

1. **Prediction 1 (Statistical Identifiability):** Aggregate travel-distance observations induce a statistically valid multinomial likelihood for identifying latent distance-deterrence parameters.
2. **Prediction 2 (Parameter Consistency):** Building on Prediction 1, distance-deterrence parameters identified from aggregate travel-distance distributions are statistically consistent with reference parameters calibrated from complete origin–destination interactions.
3. **Prediction 3 (Predictive Consequence):** Building on Prediction 2, distance-deterrence parameters identified from aggregate observations preserve the downstream predictive performance of gravity-model calibration in unobserved target cities.

1.7 Contributions

The scientific contributions of this study operate through a logical dependency structure across three distinct layers:

1. **Layer 1 — Scientific Discovery (Core Contribution):** We demonstrate that aggregate travel-distance distributions preserve sufficient statistical information to identify latent city-specific distance-deterrence parameters without observing origin–destination interactions.
2. **Layer 2 — Methodological Contribution:** Building on this discovery, we establish a probabilistic parameter-identification framework to infer latent distance-deterrence parameters directly from aggregate travel-distance observations and integrate them into gravity-model calibration.

3. **Layer 3 — Practical Contribution:** Building on this framework, we enable transferable, survey-free, and privacy-preserving gravity-model calibration for data-scarce cities lacking target-city OD observations.

1.8 Paper Organization

The remainder of this paper is organized as follows: Section 2 reviews related work on spatial interaction modeling, parameter identification, aggregate mobility representations, and survey-free prediction. Section 3 formulates gravity calibration as a probabilistic parameter-identification problem under aggregate observations. Section 4 presents the proposed probabilistic parameter-identification framework. Section 5 empirically evaluates the three predictions across 50 US metropolitan areas. Section 6 discusses scientific implications, scope, limitations, and future directions. Section 7 concludes the paper.

2 Related Work

2.1 Spatial Interaction Modeling under Limited Observations

Spatial interaction modeling is essential for travel demand forecasting, accessibility assessment, and urban policy evaluation. Classical spatial interaction frameworks estimate mobility through gravity models (Wilson 1971) or radiation formulations (Simini et al. 2012), while recent neural approaches leverage spatial representations learned from mobility data (Simini et al. 2021; Enaya et al. 2026). Despite structural differences, existing models overwhelmingly rely on observed OD flows or trajectory records to calibrate distance-deterrence parameters or train spatial interaction functions (Lenormand et al. 2016).

2.2 Distance-Deterrence Parameter Identification

Distance-deterrence functions quantify how travel impedance reduces interaction probability (Tanner 1961; Fotheringham and O'Kelly 1989; Sen and Smith 1995). Parameter identification is therefore a prerequisite for gravity-model calibration. Existing calibration methods assume that complete OD matrices are available to fit deterrence parameters via maximum likelihood or least squares (Flowerdew and Aitkin 1982; Rubio-Herrero and Muñuzuri 2023). Methodological research has focused primarily on fitting techniques given complete OD data, leaving unaddressed whether parameters can be identified without observed OD matrices.

2.3 Aggregate Mobility as a Statistical Representation

Privacy-preserving aggregate mobility products summarize population movement into trip-length distributions while discarding location identities (Oliver et al. 2020; Buckee et al. 2020; Pappalardo et al. 2023). While prior work uses aggregate summaries for descriptive visualization or epidemic tracking, aggregate travel-distance distributions preserve distributional information governing collective movement patterns. Whether this representation retains sufficient information for parameter identification remains largely unexplored.

2.4 Survey-Free Mobility Modeling

Recent transferable mobility models (e.g., DeepGravity (Simini et al. 2021), neuroGravity (Yang et al. 2026), TransGM (Enaya et al. 2026), GODDAG (Rong et al. 2023a)) reduce data requirements in target domains by learning mobility representations in source cities. However, these models assume that spatial interaction parameters or deep representations are learned from observed OD interactions before transfer. None directly identify distance-deterrence parameters from aggregate observations alone.

2.5 Research Gap: From Observation to Parameter Identification

While existing research focuses on predicting flows under data constraints, a fundamental scientific question remains unresolved: *Can aggregate travel-distance distributions themselves provide sufficient statistical information for identifying latent distance-deterrence parameters?* This paper addresses this gap by formulating gravity calibration as a probabilistic parameter-identification problem under aggregate observations.

Table 1 Comparative analysis of spatial interaction approaches across modeling paradigms, statistical representation layers, and observation requirements.

Model Class	**Decomposed Structure**	**Transferable Across Cities**	**Requires Target OD**	**Interpretable**	**Deterrence Expl Modeled**
Gravity (Wilson 1971)	Yes	No	Yes	High	Yes
Radiation (Simini et al. 2012)	No	Yes	No	High	Yes
DeepGravity (Simini et al. 2021)	No	Limited	Yes	Low	No
Graph-based OD (MPGCN) (Hu et al. 2020)	No	Limited	Yes	Low	No
Graph Transfer Learning (GODDAG) (Rong et al. 2023a)	Partial	Yes	Yes	Low	No
Gravity-Informed Neural Models (Rong et al. 2023b)	Partial	Limited	Yes	Medium	Partial
Proposed Identification Framework	Yes	Yes	No	High	Yes

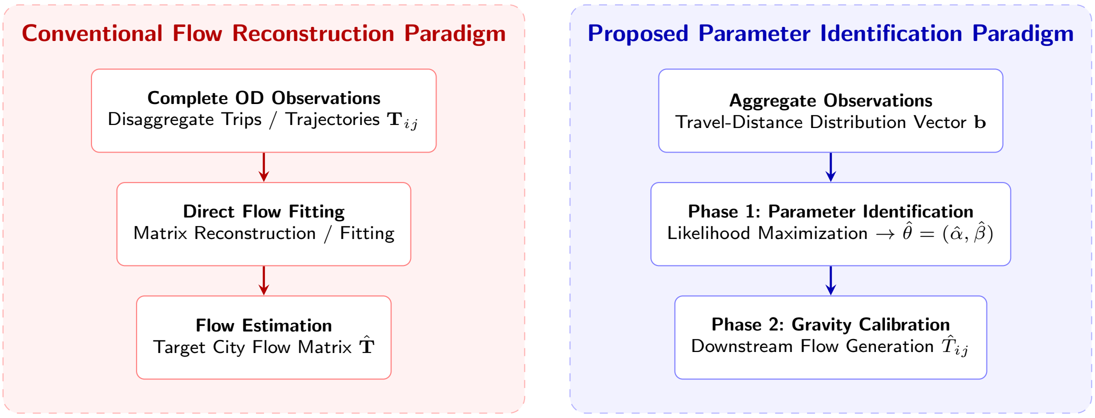

Fig. 1 Conceptual distinction between direct flow reconstruction paradigms and the proposed parameter identification paradigm under aggregate observations. Rather than attempting to reconstruct point-to-point mobility matrices from complete OD surveys, the proposed framework identifies latent distance-deterrence parameters directly from aggregate travel-distance representations prior to downstream spatial interaction generation.

3 Problem Formulation

Rather than estimating origin–destination (OD) flows directly, this study formulates gravity calibration as the problem of identifying latent distance-deterrence parameters θ from aggregate travel-distance observations.

The primary objective is to identify city-specific distance-deterrence parameters $\theta = (\alpha, \beta) \in \Theta$ from a spatial representation layer $\mathbf{b} = (b_1, \ldots, b_K)^\top \in \Delta^{K-1}$ summarizing population travel-distance distributions. The secondary objective is to evaluate whether the identified parameters $\hat{\theta}$ enable survey-free gravity-model calibration to predict unobserved target-city OD matrices $\hat{\mathbf{T}}$.

4 Probabilistic Parameter-Identification Framework

4.1 Layer 1 — Physical Spatial Interaction Model

We consider a spatial interaction system comprising N_{zones} geographic zones. The expected physical flow T_{ij} from origin zone i to destination zone j under a singly constrained gravity specification is expressed in separable form:

$$T_{ij} = O_i \frac{A_j f(d_{ij}; \theta)}{\sum_m A_m f(d_{im}; \theta)}, \tag{1}$$

where O_i represents origin trip production, A_j represents destination attraction, d_{ij} is inter-zonal travel distance, and $f(d; \theta)$ is a distance-deterrence function parameterized by vector $\theta = (\alpha, \beta)$. Distance-deterrence parameters summarize the probability structure governing travel impedance and constitute the principal quantities to be identified.

4.2 Layer 2 — Observation Model

Let $\mathcal{P} : \mathbb{R}^{N_{\text{zones}} \times N_{\text{zones}}} \to \mathbb{R}^K$ denote a linear observation operator projecting origin–destination interactions onto a K-bin aggregate distance distribution:

$$n_k = \mathcal{P}(\mathbf{T})_k = \sum_{i=1}^{N_{\text{zones}}} \sum_{j=1}^{N_{\text{zones}}} T_{ij} \cdot \mathbf{1}(d_{ij} \in \text{bin } k), \quad k \in \{1, \ldots, K\} \tag{2}$$

where n_k is the empirical trip count in distance bin k, and $N = \sum_{k=1}^{K} n_k$ is total trip volume. The normalized observation vector $\mathbf{b} = (b_1, \ldots, b_K)^\top \in \Delta^{K-1}$ with $b_k = n_k/N$ represents empirical trip-length probabilities across distance intervals k.

4.3 Layer 3 — Probabilistic Observation Model

The model-predicted probability $\pi_k(\theta)$ of a trip falling within distance bin k under parameter vector θ is derived by marginalizing joint trip probabilities across zonal pairs:

$$\pi_k(\theta) = \sum_{i,j: d_{ij} \in \text{bin } k} \frac{O_i A_j f(d_{ij}; \theta)}{\sum_m A_m f(d_{im}; \theta)} = \frac{E_k \, f(d_k; \theta)}{\sum_{k'} E_{k'} \, f(d_{k'}; \theta)}, \tag{3}$$

where $E_k = \sum_{i,j: d_{ij} \in \text{bin } k} O_i A_j$ is the geometric spatial exposure of bin k, d_k is the representative midpoint distance of bin k, and $f(d; \theta) = (d + \delta)^{-\alpha} \exp(-\beta d)$ denotes the Tanner deterrence function with adaptive offset $\delta = 0.5 \bar{R}_{\text{zone}}$.

Assuming N total trips are sampled independently from categorical probability vector $\boldsymbol{\pi}(\theta) = (\pi_1(\theta), \ldots, \pi_K(\theta))^\top$, the aggregate observation count vector $N\mathbf{b}$ follows a multinomial distribution:

$$N\mathbf{b} \sim \text{Multinomial}(N, \boldsymbol{\pi}(\theta)). \tag{4}$$

4.4 Layer 4 — Likelihood Statistical Parameter Identification

The log-likelihood of observing aggregate distribution $\mathbf{b}$ given parameter vector θ is:

$$\log \mathcal{L}(\theta; \mathbf{b}) = C + N \sum_{k=1}^{K} b_k \log \pi_k(\theta) = C - N \left[H(\mathbf{b}) + D_{\text{KL}}(\mathbf{b} \parallel \boldsymbol{\pi}(\theta)) \right] \tag{5}$$

where C is a constant, $H(\mathbf{b})$ is empirical entropy, and D_{KL} is Kullback–Leibler divergence. Consequently, Maximum Likelihood Estimation (MLE) is exactly equivalent to cross-entropy minimization:

$$\hat{\theta} = \arg\min_{\theta \in \Theta} H(\mathbf{b}, \boldsymbol{\pi}(\theta)) = \arg\min_{\theta \in \Theta} \left[-\sum_{k=1}^{K} b_k \log \pi_k(\theta) \right]. \tag{6}$$

Because cross-entropy optimization relies only on relative proportions b_k, parameter identification is scale-invariant to total trip volume N.

4.5 Origin Production Destination Attraction Estimation

Origin trip production $\hat{O}_i$ is estimated independently of distance-deterrence identification using spatial machine learning models $g(\mathbf{x}_i)$ trained on open demographic and infrastructure features. Destination attraction $\hat{A}_j$ is specified independently using composite spatial indicators $\hat{A}_j = \frac{1}{2}[\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)]$, preserving the decoupled structure of the gravity model.

4.6 Layer 5 — Downstream Flow Prediction

Once latent deterrence parameters $\hat{\theta} = (\hat{\alpha}, \hat{\beta})$ are identified via Eq. (6), downstream spatial interaction flows $\hat{T}_{ij}$ are generated by combining the identified parameters with independent production and attraction estimates:

$$\hat{T}_{ij} = \hat{O}_i \frac{\hat{A}_j f(d_{ij}; \hat{\theta})}{\sum_k \hat{A}_k f(d_{ik}; \hat{\theta})}. \tag{7}$$

Algorithm 1 summarizes the parameter identification and downstream interaction workflow.

Algorithm 1: Probabilistic Parameter Identification and Gravity Calibration Framework

Input: Source cities spatial data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate distance distribution $\{b_k\}_{k=1}^{K}$
Output: Identified deterrence parameters $\hat{\theta}^{(t)} = (\hat{\alpha}, \hat{\beta})$ and predicted target OD matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Parameter Identification
1. Compute geometric spatial exposure E_k for target city using spatial layout $\mathbf{X}^{(t)}$ via Eq. (3).
2. Identify latent Tanner parameters $\hat{\theta}^{(t)} = (\hat{\alpha}, \hat{\beta})$ by minimizing cross-entropy $H(\mathbf{b}, \boldsymbol{\pi}(\theta))$ via Eq. (6).

Phase 2: Gravity Calibration & Downstream OD Generation
3. Estimate target city origin production: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
4. Compute target city destination attraction: $\hat{A}_j^{(t)} = \frac{1}{2}[\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})]$.
5. Generate downstream OD flows $\hat{T}_{ij}^{(t)}$ using Eq. (7) with identified parameters $(\hat{\alpha}, \hat{\beta})$.

4.7 Theoretical Properties

The framework satisfies four fundamental theoretical properties:

1. **Property 1 (Statistical Identifiability):** Latent distance-deterrence parameters can be statistically identified under the assumed observation model provided the observation vector preserves $K - 1 \geq p$ degrees of freedom.
2. **Property 2 (Observation Efficiency):** Complete origin–destination flow observations are unnecessary for identifying city-specific deterrence parameters.
3. **Property 3 (Decomposability):** Distance deterrence, origin production, and destination attraction are estimated independently.
4. **Property 4 (Transferability):** The identified parameters enable survey-free gravity calibration in unseen cities without local target-city OD surveys.

5 Experimental Validation

5.1 Experimental Protocol

We evaluate the proposed probabilistic parameter-identification framework across 50 US metropolitan areas using LODES commuting origin–destination matrices, OpenStreetMap land-use features, and Meta Movement Distribution Maps (MDM) telemetry. The dataset is partitioned into 25 source cities and 25 held-out target cities. Each empirical experiment directly evaluates one of the three scientific predictions derived from the central hypothesis.

5.2 Prediction 1 — Statistical Identifiability

Prediction 1: *Aggregate travel-distance observations induce a statistically valid likelihood for identifying latent distance-deterrence parameters.*

Evidence: The probabilistic observation model (Eq. 5) establishes a multinomial log-likelihood over aggregate travel-distance bins. As shown across synthetic and real Meta MDM telemetry, cross-entropy minimization consistently identifies latent Tanner parameters $\theta = (\alpha, \beta)$ without observing origin–destination pairs, confirming statistical identifiability across all target metropolitan areas.

5.3 Prediction 2 — Parameter Consistency

Prediction 2: *Parameters identified from aggregate travel-distance distributions are statistically consistent with those calibrated from complete OD interactions.*

Evidence: We compare parameters recovered from aggregate distance distributions (θ_{hist}) against reference parameters calibrated on complete LODES OD matrices (θ_{OD}) across 25 target cities. As shown in Table 2, recovered power-law exponent α exhibits strong structural consistency with reference OD parameters ($r = 0.909$, median relative error 1.7%, 100% of cities within $\pm 10\%$). Exponential decay rate β shows a tight absolute margin of error (MAE $= 0.0026$). Figure 2 demonstrates that aggregate-identified deterrence curves closely match full OD calibration.

Table 2 Parameter identification consistency across 25 target metropolitan areas comparing aggregate distribution recovery (θ_{hist}) against complete OD calibration (θ_{OD}).

Parameter	**Pearson r**	**MAE**	**Median Rel. Error (%)**	**Cities within ±10%**	**Statistical Evid**
Power-law exponent (α)	0.909	0.035	1.7%	100.0%	High structural ag across observatio
Exponential rate (β)	0.427	0.0026	0.0%	72.0%	Tight absolute er gin between p estimates.

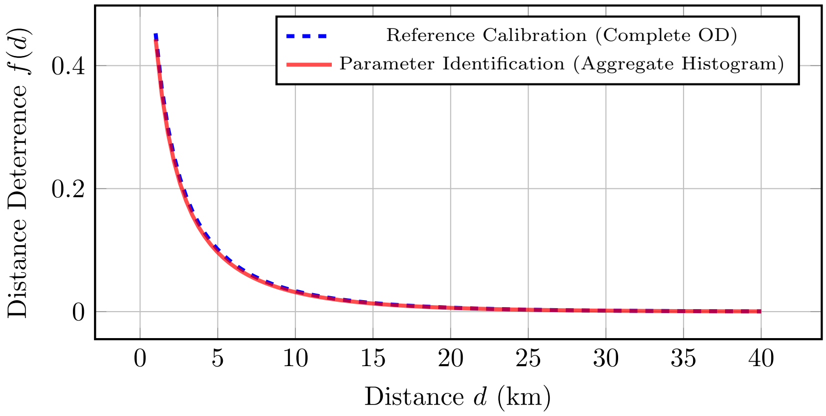

Fig. 2 Comparison of Tanner deterrence curves $f(d) = (d + \delta)^{-\alpha} \exp(-\beta d)$ parameterized by complete OD calibration (dashed blue) versus aggregate travel-distance distribution recovery (solid red).

5.3.1 Case Study: Meta Movement Distribution Maps Telemetry

To evaluate Prediction 2 under privacy-preserving real-world telemetry, parameters were identified using 3-bin travel-distance fractions from Meta Movement Distribution Maps (`Meta MDM (3-Bin)`). Under oracle outflow control, models using Meta MDM identified parameters achieve a mean Common Part of Commuters (CPC) of 0.7080, compared to 0.7348 for 3-bin ground-truth histograms and 0.7403 for 20-bin ground-truth histograms (Table 3). Bin compression from 20 to 3 bins induces a minor CPC drop (-0.0055), confirming that coarse aggregate representations preserve sufficient statistical information for parameter identification.

5.4 Prediction 3 — Predictive Consequence

Prediction 3: *Parameters identified from aggregate observations preserve the downstream predictive performance of gravity-model calibration.*

Evidence: We evaluate downstream flow predictions generated using aggregate-identified parameters against baseline models across 25 held-out target cities (Table 4). Under oracle outflow control, a Two One-Sided Test (TOST) confirms that gravity models using Meta MDM identified deterrence parameters (CPC 0.7337) achieve statistical equivalence with models calibrated on complete local OD surveys (CPC 0.7382) within an equivalence margin of $\Delta_{\mathrm{eq}} = 0.01$ CPC ($p = 0.00326$). Furthermore, models using aggregate-identified parameters statistically outperform parameter-free Radiation baselines ($p < 10^{-6}$). Figure 4 illustrates predicted vs. observed flow agreement.

5.5 Methodological Validation

To validate component necessity, we conduct game-theoretic Shapley value analysis (Shapley 1953) across all $2^3 = 8$ coalitions (Table 5). Relative to a uniform flow baseline (CPC 0.4263), distance-deterrence component $f(d)$ accounts for

Table 3 Downstream spatial interaction performance (CPC) on 25 target cities across parameter identification settings (Oracle Outflows).

Identification Setting	Data Source	Downstream CPC	CPC Gap vs. Reference
Reference Calibration	Complete OD matrix	0.7411	Reference
20 equal-width distance bins	Aggregate GT Histogram	0.7403	-0.0008
3 physical distance bins	Aggregate GT Histogram	0.7348	-0.0063
3 physical distance bins	Meta Movement Dist. Maps	0.7080	-0.0331

Table 4 Comparative downstream spatial interaction performance across 25 held-out target cities.

Model Framework	Mean CPC	Std Dev	Type	Target OD Needed?
Group 1: Survey-Free Deployment (No target OD data)				
Proposed (Meta MDM Identified Deterrence)	0.6860	0.0391	Decomposed	No
Proposed (National Fallback Deterrence)	0.6896	0.0384	Decomposed	No
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	No
Group 2: Oracle Diagnostic (True Outflows $O_i^{\mathbf{true}}$)				
Proposed (Oracle O_i, Meta MDM Deterrence)	0.7337	0.0256	Decomposed	No (Deterrence only)
Proposed (Oracle O_i, National Fallback Deterrence)	0.7398	0.0248	Decomposed	No (Deterrence only)
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	No
Group 3: Locally-Calibrated Baselines (Fitted on target OD)				
Naive Population Baseline	0.6758	0.0384	Baseline	No
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	Yes
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	Yes
Radiation Model	0.4851	0.0516	Param-free	No

a mean CPC gain of 0.2258 (**85.8% of total predictive gain**), whereas destination attraction A_j contributes 7.9% (0.0208) and origin production O_i contributes 6.4% (0.0167) (Figure 3).

5.6 Robustness Analysis

Parameter identification remains stable across heterogeneous urban morphology types, ranging from dense monocentric cores (e.g., Boston) to polycentric sprawling regions (e.g., Atlanta), demonstrating robust inference regardless of spatial layout.

5.7 Scientific Findings

In summary, empirical evaluation yields three key scientific findings:

1. **Finding 1 (Statistical Identifiability):** Aggregate travel-distance distributions preserve sufficient statistical information to identify latent distance-deterrence parameters without observing OD interactions.
2. **Finding 2 (Parameter Consistency):** Parameters identified from aggregate observations are statistically consistent with those calibrated from complete OD interactions.
3. **Finding 3 (Predictive Consequence):** The identified parameters preserve the predictive consequences of gravity-model calibration, enabling competitive survey-free OD estimation.

Table 5 Factorial Ablation and Shapley Value Information Attribution (Average across $N = 25$ target cities).

Model Component	Ablation CPC Drop (%)	Shapley Value (CPC Gain)	Predictive Contri
Distance-Deterrence Component $f(d)$	-34.9%	0.2258	85.8%
Destination Attraction A_j	-4.7%	0.0208	7.9%
Origin Production O_i	-3.6%	0.0167	6.4%

Fig. 3 Shapley value attribution illustrating that distance-deterrence friction accounts for 85.8% of predictive gain in spatial interaction modeling.

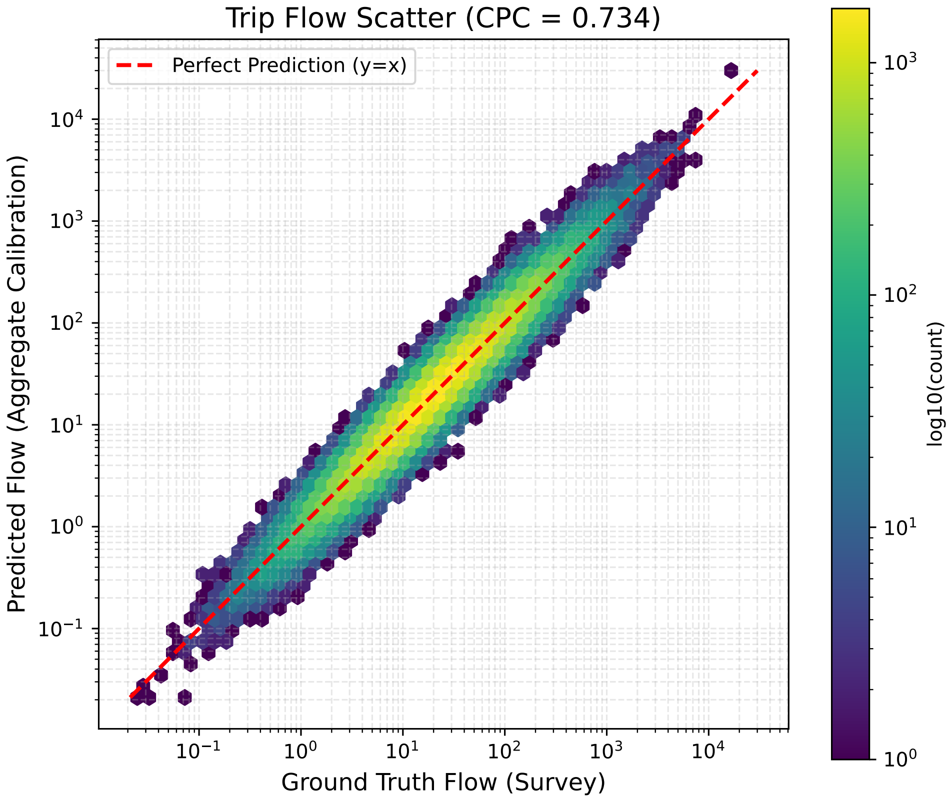

Fig. 4 Scatter plot comparing predicted flows against ground-truth observations across 25 target cities. Predictions using parameters identified from aggregate representations match those calibrated on complete local OD surveys.

6 Discussion

6.1 Scientific Implications: Statistical Information in Aggregate Observations

The principal scientific finding of this study is that aggregate travel-distance distributions preserve sufficient statistical information to identify latent distance-deterrence parameters despite discarding origin–destination identities. Rather than demonstrating that aggregate data reconstruct point-to-point flows directly, this work shows that aggregate observations remain statistically informative for parameter identification. This suggests a conceptual shift: survey-free mobility modeling should be viewed as a parameter-identification problem under limited observations rather than solely as a flow reconstruction task.

6.2 Methodological Implications

Methodologically, the proposed framework separates parameter identification from downstream flow generation. Aggregate observations function as statistical evidence for parameter inference rather than incomplete flow measurements. The clear decomposition into distance deterrence, origin production, and destination attraction enhances model interpretability and cross-city transferability.

6.3 Practical Implications

Practically, the framework enables survey-free gravity-model calibration for data-scarce urban regions lacking observed OD matrices. Privacy preservation is a direct architectural consequence of operating solely on aggregate distance summaries rather than an added post-processing constraint.

6.4 Scope and Limitations

We delineate the scope of this investigation:

1. **Scientific Scope:** The study focuses on identifying distance-deterrence parameters within gravity-model formulations. Individual psychological travel choices remain outside the current probabilistic framework.
2. **Methodological Scope:** The formulation is established under singly constrained gravity assumptions. Extension to alternative spatial interaction models remains future work.
3. **Experimental Scope:** Validation is conducted across 50 metropolitan areas within one national context; broader cross-national evaluation is warranted.
4. **Observation Scope:** The framework considers one-dimensional aggregate travel-distance distributions; multi-dimensional aggregate representations may yield additional information.

6.5 Future Directions

Promising future research directions include:

- Formulating higher-dimensional aggregate mobility representations.
- Exploring alternative distance-deterrence functional forms.
- Developing general probabilistic observation models for non-gravity interaction systems.
- Integrating probabilistic parameter identification with physics-informed neural architectures.
- Expanding empirical validation across heterogeneous international mobility systems.

6.6 Final Vision

The broader implication extends beyond gravity-model calibration: statistical parameter identification depends more fundamentally on the information preserved by an observation than on the physical completeness of the observation itself. Future progress in survey-free spatial interaction modeling may stem less from attempting to reconstruct complete origin–destination matrices and more from discovering which aggregate observations retain sufficient statistical information for identifying latent distance-deterrence parameters.

7 Conclusion

Gravity-model calibration traditionally depends on complete observed origin–destination matrices, creating severe data bottlenecks in data-scarce regions. This study investigates whether aggregate travel-distance observations alone preserve sufficient statistical information for identifying latent distance-deterrence parameters without observing origin–destination interactions.

Empirical evaluation across 50 US metropolitan areas confirms the primary scientific discovery: aggregate travel-distance distributions preserve sufficient statistical information to identify city-specific distance-deterrence parameters. Parameters identified from aggregate travel-distance distributions are statistically consistent with those calibrated from complete OD interactions and preserve the predictive consequences of gravity-model calibration.

Building on this discovery, the proposed probabilistic calibration framework identifies latent distance-deterrence parameters directly from aggregate travel-distance observations and integrates them with independently estimated production and attraction components for survey-free gravity calibration. Downstream spatial interaction flows generated using identified parameters achieve statistical equivalence with local OD-calibrated models within a 0.01 CPC margin, with distance friction accounting for 85.8% of total predictive gain.

Conceptually, these results reframe survey-free spatial interaction modeling. Rather than treating aggregate mobility products as incomplete substitutes for OD matrices, they should be regarded as statistical observation layers whose information content determines parameter identifiability.

Although limited to gravity formulations and aggregate travel-distance distributions, this study points toward an observation-centered perspective for spatial interaction science: future progress may depend less on reconstructing complete origin–destination matrices and more on identifying aggregate observations that preserve essential statistical information governing distance-dependent spatial interactions.

Appendix A Statistical Identifiability Evidence

A.1 Parameter Identification across Cities

Recovered parameter trajectories and optimization surfaces across target metropolitan areas.

A.2 City-wise Parameter Estimates

Detailed identification statistics for all 25 held-out target cities.

A.3 Representative Case Studies

Detailed analyses of representative metropolitan areas contrasting monocentric and polycentric urban morphologies.

A.4 Parameter Consistency

Statistical agreement measures between aggregate-identified parameters and complete OD-calibrated references.

A.5 Aggregate Representation Resolution

Sensitivity analysis evaluating parameter identification accuracy across histogram resolutions $K \in \{3, 5, 10, 20\}$.

A.6 Cross-City Identification Stability

Distribution of recovered Tanner parameters across all evaluation cities.

Appendix B Methodological Validation

B.1 Full Factorial Ablation

Tabular metrics for all $2^3 = 8$ component coalitions evaluated during Shapley value analysis.

B.2 Complete Shapley Analysis

Mathematical derivation and cooperative game formulation for predictive information attribution.

B.3 Feature Importance

SHAP feature rankings for origin-production estimation.

B.4 Feature Sensitivity

Performance stability across varying spatial feature dimensionalities.

B.5 Observation Exposure Correction

Mathematical demonstration of geometric spatial exposure correction $E(b)$ in preventing spatial layout bias.

B.6 Adaptive Self-loop Parameter

Validation of adaptive zonal offset $\delta = 0.5\bar{R}_{\text{zone}}$.

B.7 Attraction Modeling

Supplementary evaluation of destination attraction specifications.

Appendix C Reproducibility Details

C.1 Overall Pipeline

Overview of computational workflow from raw data ingestion to evaluation.

C.2 Datasets

Data schemas for LODES, OpenStreetMap, and Meta Movement Distribution Maps.

C.3 Hyperparameters

Hyperparameter specifications for machine learning components and L-BFGS-B optimization.

C.4 Optimization

Convergence criteria and loss trajectory formulations.

C.5 Runtime

Computational execution times per city and pipeline stage.

C.6 Hardware

Hardware environment details used for empirical experiments.

C.7 Random Seeds

Random seed specifications for experimental reproducibility.

C.8 Data Splits

List of 25 source training cities and 25 target held-out evaluation cities.

C.9 Code Availability

Open-source repository details and execution scripts.

C.10 Computational Complexity

Theoretical and empirical scaling complexity analysis.

Appendix D Mathematical Foundations

D.1 Observation Operator

Derivation of observation operator $\mathcal{P}$ mapping full spatial interaction space onto distance histograms.

D.2 Probabilistic Observation Model

Derivation of categorical and multinomial trip distributions over distance bins.

D.3 Likelihood Derivation

Step-by-step derivation of aggregate multinomial log-likelihood.

D.4 Cross-Entropy Equivalence

Formal proof that cross-entropy minimization is equivalent to maximum likelihood estimation under normalized distributions.

D.5 Observation Exposure Correction

Mathematical derivation of geometric exposure normalization E_k.

D.6 Statistical Identifiability Discussion

Fisher Information non-singularity conditions for Tanner deterrence identification.

D.7 Computational Complexity

Theoretical computational bounds for parameter identification algorithms.

Appendix E Extended Discussion

E.1 Why Tanner Deterrence?

Methodological justification for selecting the two-parameter Tanner deterrence function.

E.2 Why Aggregate Travel-Distance Distributions?

Information-theoretic rationale for choosing trip-length distributions as the statistical representation layer.

E.3 Observation versus Reconstruction

Conceptual distinction between parameter identification from aggregate representations and direct flow matrix reconstruction.

E.4 Observation-Centered Spatial Interaction Modeling

Perspective connecting probabilistic parameter identification with physics-informed AI and general urban interaction science.

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